

Pre- & Post-Test

The pre-test and post-test will be identical and only allow one attempt each.

Participants will answer all selected questions (15 questions in total) for each test. The distribution of these questions between the pre-test and post-test is outlined below.

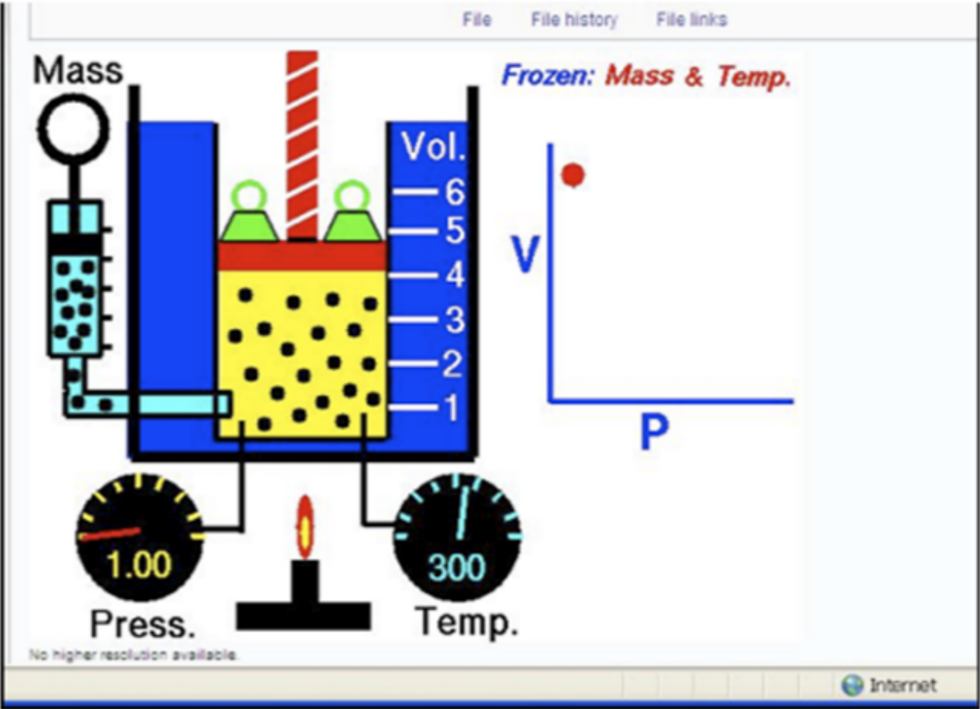
Learning Objectives (LOs)	Fact (F)	Skill (S)
Describe the multimedia principle and why it is effective for learning	2	4
Recognize when the multimedia principle has been violated and when it has been applied well	5	4

Question List

The correct answer for each question is **bolded**.

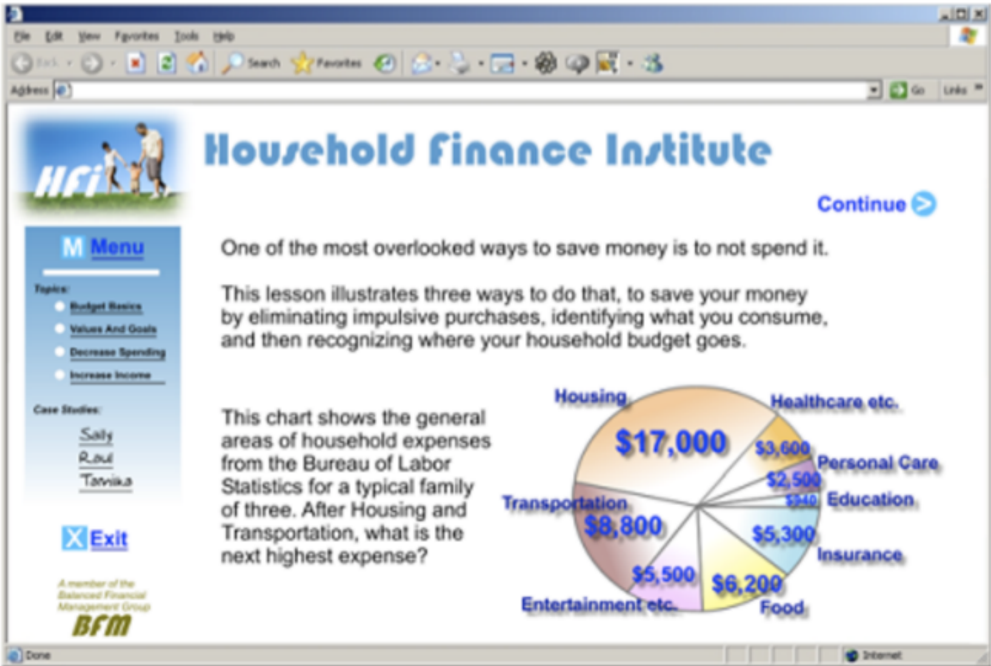
F v. S	S
LO	Describe the multimedia principle and why it is effective for learning
Prompt	A lesson on Microsoft Outlook includes a screen capture to illustrate the menus and tool bars. The screen capture is an example of a(n) _____ graphic.
Choices	A. Decorative B. Representational C. Organizational D. Relational

F v. S	S
LO	Describe the multimedia principle and why it is effective for learning

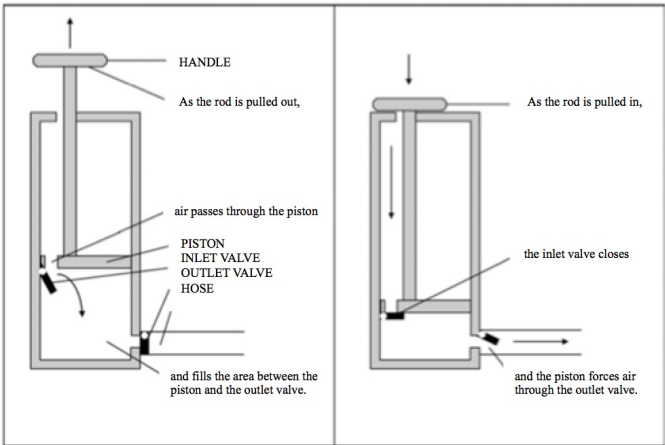
Prompt	 <i>Alt text: The image shows a visual diagram illustrating the relationship between gas volume, pressure, and temperature. On the left side, there is a cylinder with a moveable piston representing gas volume, a gauge showing pressure (labeled "Press.") at 1.00, and a temperature gauge (labeled "Temp.") at 300. The cylinder is placed over a flame. On the right side of the image, there is a graph with "V" on the vertical axis representing volume and "P" on the horizontal axis representing pressure, with a red dot plotted at an intersection. The title of the image is "Frozen: Mass & Temp."</i> The illustration above of Boyle's Law (<i>Boyle's law is a gas law that describes how the pressure of a gas tends to decrease as the volume of the container increases</i>) is an example of a(n)
Choices	A. Decorative graphic B. Organizational graphic C. Representational graphic D. Interpretive graphic

F v. S	S
LO	Describe the multimedia principle and why it is effective for learning
Prompt	According to the multimedia principle, a relevant graphic can improve learning because the learner is:
Choices	A. Actively engaged

	B. Motivated C. Focused D. Matched to a visual learning style
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F v. S	S
LO	Describe the multimedia principle and why it is effective for learning
Prompt	 <i>Alt text: The image shows a web page from the "Household Finance Institute" with a lesson about saving money by reducing impulsive purchases and recognizing household budget allocations. On the left side is a vertical menu titled "Menu" with topics such as "Budget Basics," "Values and Goals," "Decrease Spending," and "Increase Income." There are also case studies with links to "Sally," "Ron," and "Tanika."</i> <i>The main section contains text explaining the lesson's purpose and features a pie chart displaying household expenses for a typical family of three, sourced from the Bureau of Labor Statistics. The chart highlights various categories, including:</i> <i>Housing: \$17,000</i> <i>Transportation: \$8,800</i> <i>Healthcare: \$3,600</i> <i>Food: \$6,200</i> <i>Entertainment: \$5,500</i> <i>Insurance: \$5,300</i> <i>Other smaller categories like Personal Care and Education are also shown.</i> <i>A "Continue" button is located on the upper right of the screen, and an "Exit" button is at the bottom left. The page encourages users to identify the next highest expense</i>

	after Housing and Transportation. The graphic on the screen above is an example of which type of visual:
Choices	A. Decorative B. Interpretive C. Relational D. Organizational

F v. S	F
LO	Describe the multimedia principle and why it is effective for learning
Prompt	How a Bicycle Pump Works A. As the rod is pulled out, air passes through the piston and fills the area between the piston and the outlet valve. As the rod is pushed in, the inlet valve closes and the piston forces air through the outlet valve. B.  <i>Alt text: The image is titled "How a Bicycle Pump Works" and is divided into two sections, labeled A and B.</i> • <i>Section A contains a textual description of how a bicycle pump works. It explains that when the rod is pulled out, air passes through the piston and fills the area between the piston and the outlet valve. When the rod is pushed in, the inlet valve closes, and the piston forces air through the outlet valve.</i> • <i>Section B provides a visual diagram showing the inner mechanics of a bicycle pump. The diagram has labels such as "Handle," "Piston," "Inlet Valve," "Outlet Valve," and "Hose." Arrows illustrate the motion of the rod being pulled out (air passes through the piston) and pushed in (the inlet valve closes and air is forced through the outlet valve). The diagram visually corresponds to the process described in Section A.</i> Assume: 1) Learning is being measured with a post-assessment of transfer such as: If the inlet valve in a different pump remains open when the rod is pushed in, what will happen to the air pressure inside the pump?.

	2) Students are beginners -- closer to novice. Which of these e-learning cases, A or B, is better for student learning?
Choices	A. A B. B C. No difference in learning D. The answer depends on critical information that is not provided

F v. S	F
LO	Describe the multimedia principle and why it is effective for learning
Prompt	How does the principle you chose for the lesson on how a bicycle pump works facilitate better learning?
Choices	A. By increasing generative processing B. By decreasing generative processing C. By increasing extraneous processing D. By decreasing extraneous processing

F v. S	S
LO	Describe the multimedia principle and why it is effective for learning
Prompt	How a Bicycle Pump Works A. As the rod is pulled out, air passes through the piston and fills the area between the piston and the outlet valve. As the rod is pushed in, the inlet valve closes and the piston forces air through the outlet valve. B. The diagram illustrates the two-stage process of a bicycle pump. Stage A shows the rod being pulled out, with air passing through the piston and filling the space between the piston and the outlet valve. Stage B shows the rod being pushed in, with the inlet valve closing and the piston forcing air through the outlet valve. Labels include: HANDLE, PISTON, INLET VALVE, OUTLET VALVE, and HOSE. <i>Alt text: The image is titled "How a Bicycle Pump Works" and is divided into two sections, labeled A and B.</i> Section A contains a textual description of how a bicycle pump works. It explains that when the rod is pulled out, air passes through the piston and

	<i>fills the area between the piston and the outlet valve. When the rod is pushed in, the inlet valve closes, and the piston forces air through the outlet valve.</i> • <i>Section B provides a visual diagram showing the inner mechanics of a bicycle pump. The diagram has labels such as "Handle," "Piston," "Inlet Valve," "Outlet Valve," and "Hose." Arrows illustrate the motion of the rod being pulled out (air passes through the piston) and pushed in (the inlet valve closes and air is forced through the outlet valve). The diagram visually corresponds to the process described in Section A.</i> Assume: 3) Learning is being measured with a post-assessment of verbatim recall, such as: Did the text say "when the rod is pulled out, air passes through the piston"? 4) Students are beginners -- closer to novice. Which of these e-learning cases, A or B, is better for student learning?
Choices	E. A F. B G. No difference in learning H. The answer depends on critical information that is not provided

F v. S	S
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	According to the multimedia principle, learning is better when a lesson:
Choices	A. Is delivered on a computer B. Adds graphics to a text description C. Adds text to a graphic illustration D. Uses animation rather than still visuals

F v. S	S
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	Relevant graphics added to words improve learning most when:
Choices	A. The content is factual B. The lesson is not interesting C. The delivery medium is a computer D. The learners are new to the content

F v. S	S
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	Animated graphics are most useful to teach:
Choices	A. Procedures such as tying a note or first aid techniques B. Processes such as how lightning works or how a bike pump works C. Concepts such as a braking system or how ocean waves work D. Facts

F v. S	S
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	Based on current evidence, which lesson version would most help learners build a deeper understanding of how lightning forms:
Choices	A. A clear text description alone B. A clear text description accompanied by static visuals C. A clear text description accompanied by animated visuals D. An animation without words

F v. S	F
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	Fill in the sentences below about the application of the multimedia principle. <i>A history lesson on the original thirteen colonies with a graphic of The British flag to represent America's separation from Great Britain.</i> The lesson described above is a [Dropdown 1] of the multimedia principle because it uses a [Dropdown 2] image. A [Dropdown 3] graphic would be appropriate for this lesson.
Choices	Dropdown 1: A. good application B. violation Dropdown 2: A. relevant B. decorative Dropdown 3: A. representational or organizational

	B. transformational or interpretive
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F v. S	F
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	<i>A chemistry lesson on the six inert gases with an image of an explosion resulting from a chemical reaction of a gas under pressure.</i> The lesson described above is a [Dropdown 1] of the multimedia principle because it uses a [Dropdown 2] image. A [Dropdown 3] graphic would be appropriate for this lesson.
Choices	Dropdown 1: A. good application B. violation Dropdown 2: A. relevant B. decorative Dropdown 3: A. representational or organizational B. transformational or interpretive

F v. S	F
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	<i>An anatomy lesson on locating and naming the bones of the human body that includes an image of the human skeleton.</i> The lesson described above is a [Dropdown 1] of the multimedia principle because it uses a [Dropdown 2] image. A [Dropdown 3] graphic would be appropriate for this lesson.
Choices	Dropdown 1: A. good application B. violation Dropdown 2: A. relevant B. decorative Dropdown 3: A. representational or organizational B. transformational or interpretive

F v. S	F
LO	Recognize when the multimedia principle has been violated and when it has been applied well
Prompt	<i>A lesson on blood flow through the heart includes a cross-section graphic of a heart showing the chambers and ventricles with directional arrows.</i> The lesson described above is a [Dropdown 1] of the multimedia principle because it uses a [Dropdown 2] image. A [Dropdown 3] graphic would be appropriate for this lesson.
Choices	Dropdown 1: A. good application B. violation Dropdown 2: A. relevant B. decorative Dropdown 3: A. representational or organizational B. transformational or interpretive

For pre-test and post-test, one attention check question for each is added:

- [Pre-Test Attention Check] This is a test to see if you are carefully reading the questions.
 - This is the wrong answer.
 - Even though this answer might look correct, do not select it.
 - This is the correct answer; select this one. [correct answer]
 - If you are reading carefully, you will not select this option.
- [Post-Test Attention Check] Please select the second answer choice listed below, as this is an attention-check question.
 - The first answer choice, which should not be selected.
 - The second answer choice, which is the one you need to pick. [correct answer]
 - The third answer choice, which is incorrect.
 - The fourth answer choice, which is a distraction.